

Attention Beyond the Training Length

How Score Scaling Controls Length Generalization in a Reduced Binary Attention Classifier

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Motivation

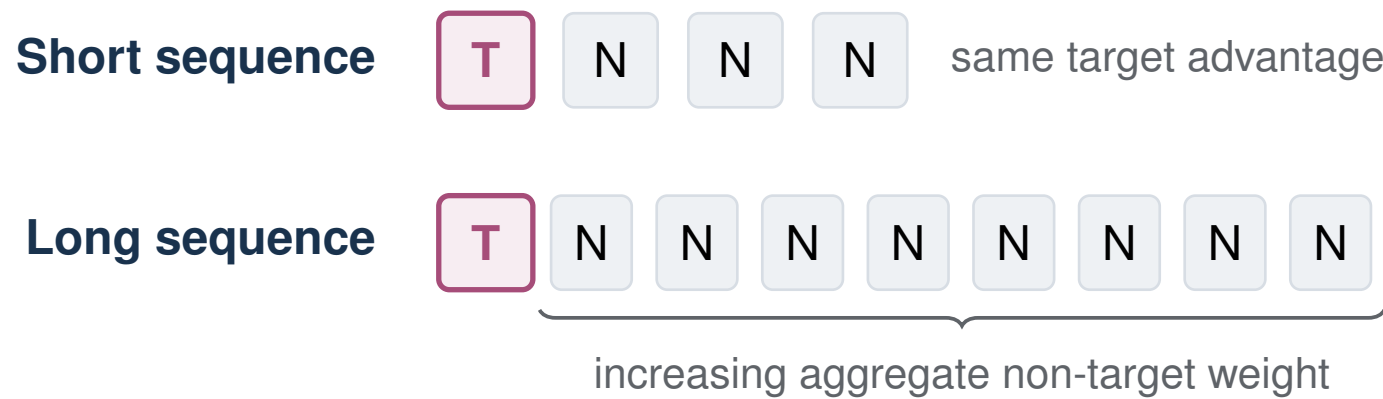
Training on short sequences does not guarantee reliable performance on much longer inputs [2]. As sequence length grows, the target must compete with more non-target tokens for the same attention budget.

Research Question

How must attention sharpen with length to remain concentrated on the target?

Why Fixed Attention Dilutes with Length

The model detects whether a designated target token is present in the sequence.



The target retains an advantage over every individual non-target, but the non-targets grow in number and dominate in aggregate.

Let $p_t(n)$ be the target's share of attention. With target score a and $n - 1$ identical non-target scores b , standard softmax attention [1] reduces to

$$p_t(n) = \frac{\exp(a)}{\exp(a) + (n - 1) \exp(b)} = \frac{1}{1 + (n - 1) \exp(-\Delta)}$$

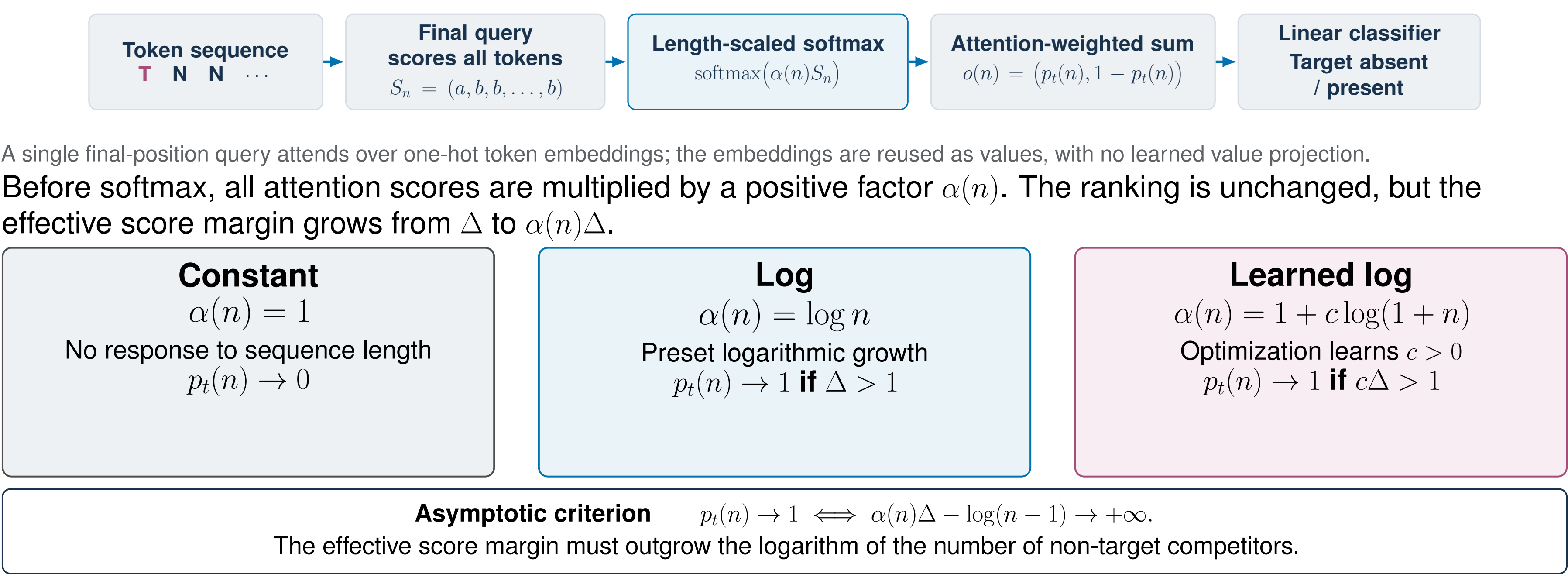
$\Delta = a - b > 0$.

For any fixed target–non-target score margin Δ , $p_t(n) \rightarrow 0$ as the number of competitors grows without bound.

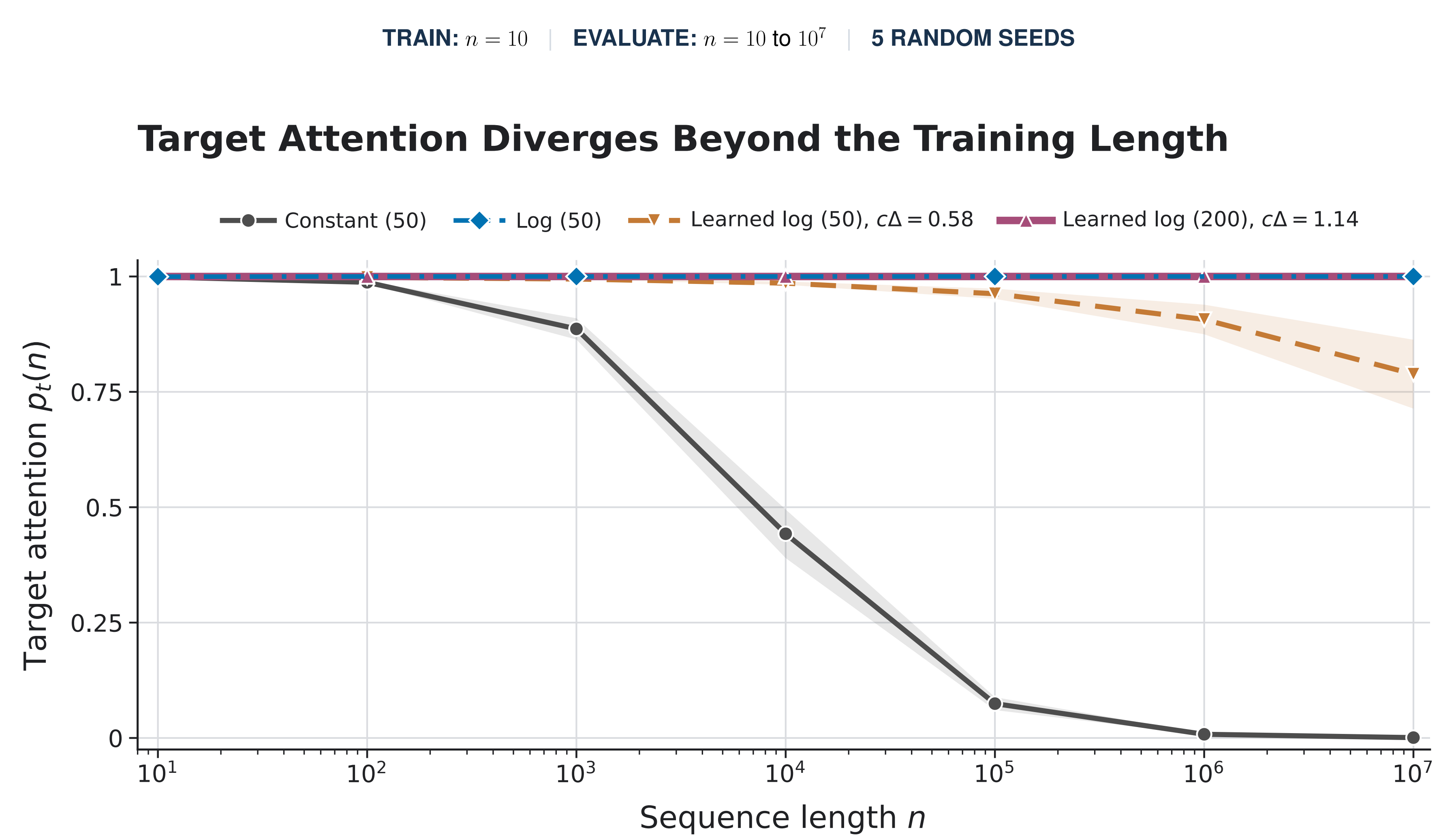
A fixed score margin separates the target from each non-target, but cannot offset their growing aggregate weight.

What if the effective score margin grows with sequence length?

How Length-Aware Scaling Counters Dilution



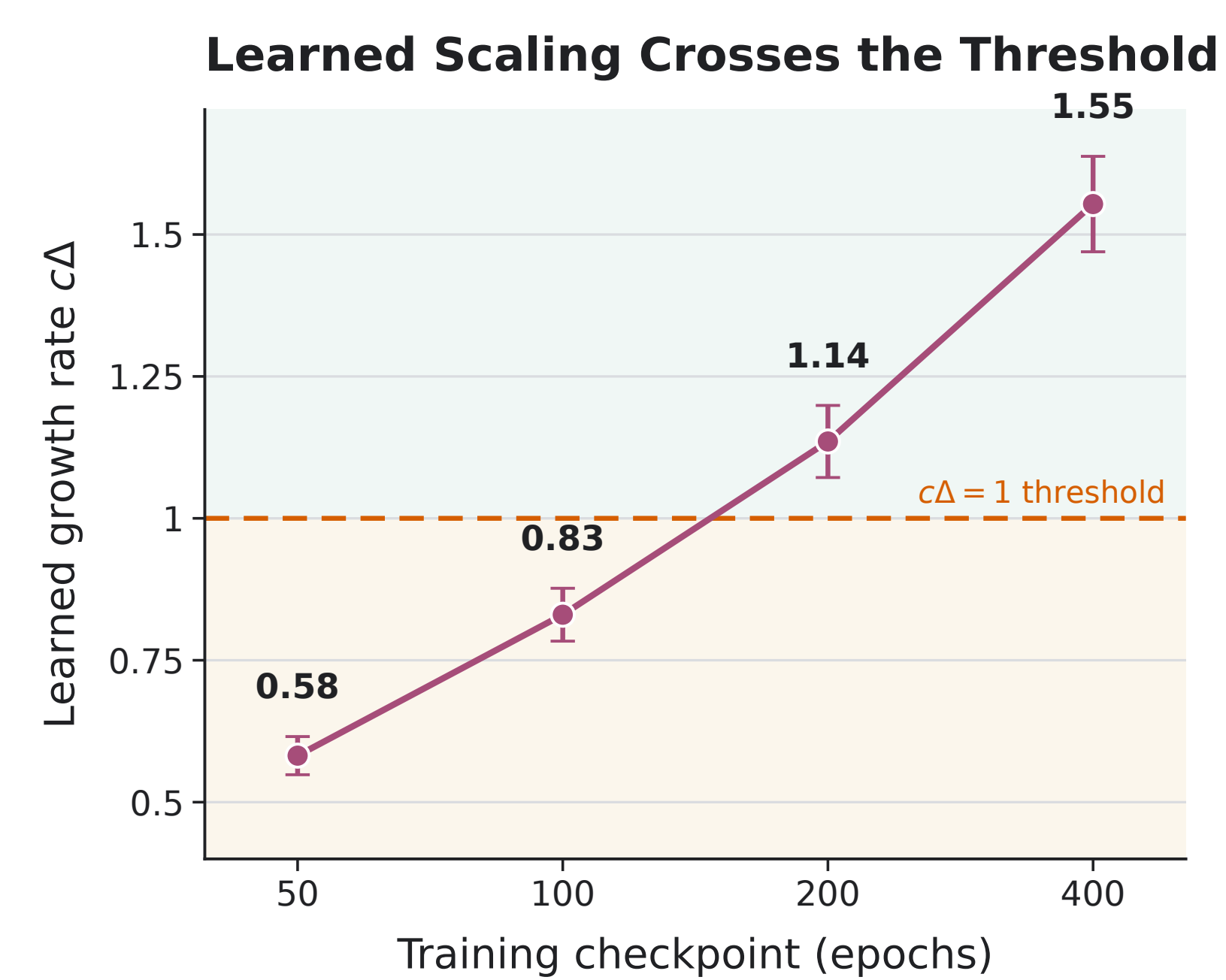
Same Training-Length Accuracy. Sharply Different Long-Length Behavior.



Representative runs evaluated from the training length to 10^7 . Means over five seeds; bands show ± 1 s.d.

Same Margin, Different Outcome. Constant (50) and Learned log (200) both learn $\Delta \approx 9$, yet their positive accuracy at $n = 10^7$ is 0% versus 100%. This comparison isolates length scaling from the raw learned score margin.

Learned-Log Threshold



The learned growth rate crosses $c\Delta = 1$ between the 100- and 200-epoch checkpoints. Points show means over five seeds; error bars show ± 1 s.d.

A Finite Pass Can Hide Later Failure. Every learned-log checkpoint reaches 100% accuracy at $n = 10^7$, but only the 200- and 400-epoch checkpoints satisfy $c\Delta > 1$. The earlier checkpoints are predicted to fail at still greater lengths.

Key Takeaways & Scope

1 Fixed scaling postpones, but cannot prevent, target-attention dilution.

Any fixed target–non-target score margin is eventually overwhelmed by the growing aggregate non-target weight.

2 Sufficiently strong logarithmic sharpening can prevent dilution.

Target attention converges to one when $\Delta > 1$ for Log or $c\Delta > 1$ for Learned log.

Scope. These thresholds are exact for the reduced two-score classifier studied here; whether analogous conditions hold in full transformers remains open.

References & Acknowledgments

[1] Vaswani et al. (2017). *Attention Is All You Need*. NeurIPS.
[2] Press et al. (2022). *Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation*. ICLR.

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